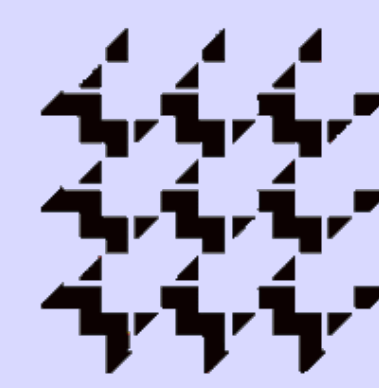




Circle Packing and the Discrete Schwarz Lemma



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Introduction

The study of circle packings provides a discrete analog to classical complex analysis. By assigning radii to vertices of a triangulated surface, we can define geometric structures like edge lengths and curvature. This framework allows us to explore discrete versions of fundamental theorems like the Schwarz Lemma.

Core Concepts

Triangulated Surface $(S, \mathcal{T})$: A surface S decomposed into a set of triangles $\mathcal{T}$.

Circle Packing Metric: A radius function $r : V \rightarrow (0, \infty)$ where the induced edge lengths satisfy the triangle inequality for every face in $\mathcal{T}$. For tangent circles, edge lengths are $l_{ij} = r_i + r_j$.

Discrete Curvature: For a vertex v_i , the curvature is the angle defect:

$$K_r(v_i) := 2\pi - \sum_{ijk \in F} \theta_i^{jk}$$

where θ_i^{jk} is the angle at vertex v_i in the triangle (i, j, k) .



Figure: Discrete curvature as angle defect.

Previous Work: Rigidity and the Schwarz Lemma

Thurston's Rigidity Theorem (1985): If two circle packing metrics R and r on a closed triangulated surface have the same discrete curvature ($K_R = K_r$), then they are related by a constant scaling factor ($R = \lambda r$).

Discrete Schwarz-Pick Lemma (Beardon Stephenson, 1991): Let $(S, \mathcal{T})$ be a closed triangulated surface with vertices partitioned $V = V_1 \sqcup V_2$. If R and r are circle packing metrics such that:

Generalization: Intersection Angles

We generalize the classical circle packing model by allowing adjacent circles to intersect at a prescribed angle, rather than just being tangent.

Intersection Angles Φ : A function $\Phi : E \rightarrow [0, \pi)$ that assigns an intersection angle to each edge.

The edge length l_{ij} is now given by the law of cosines:

$$l_{ij}^2 = r_i^2 + r_j^2 + 2r_i r_j \cos(\Phi_{ij})$$

Main Theorem 1: Discrete Schwarz Lemma for Intersection Angles

Let $(S, \mathcal{T}, \Phi)$ be a closed triangulated surface with intersection angles. Let the vertices be partitioned $V = V_1 \sqcup V_2$.

If for every triangle $\triangle v_i v_j v_k$ in the triangulation, the following condition holds for any permutation $\{a, b, c\}$ of $\{i, j, k\}$:

$$\cos(\Phi_a) + \cos(\Phi_b) \cos(\Phi_c) \geq 0$$

And if R and r are circle packing metrics such that:

- $K_R(v) \geq K_r(v)$ for all $v \in V_1$
- $R(v) \geq r(v)$ for all $v \in V_2$

Then $R(v) \geq r(v)$ for all $v \in V$.

Proof Sketch

The proof follows the structure of the original discrete Schwarz lemma but requires re-evaluating the properties of the discrete curvature map. The main challenge is to show that the Jacobian of the curvature map remains an M-matrix under the new condition on intersection angles, which ensures the comparison principle holds.

Further Generalization: Inversive Distances

A further generalization uses inversive distance $l_{ij} \in [1, \infty)$ to define the relationship between circles. The edge length is given by:

$$l_{ij}^2 = r_i^2 + r_j^2 + 2r_i r_j l_{ij}$$

When $l_{ij} = 1$, the circles are tangent. When $l_{ij} > 1$, the circles are separated.

Conjecture: The Discrete Schwarz Lemma holds for any inversive distances.

Counterexample for Inversive Distances

The conjecture is false. We constructed a counterexample on a simple 4-vertex triangulation. Let $V_1 = \{v\}$ and $V_2 = \{a, b, c\}$. We define two circle packing metrics, r and R .



Figure: The triangulation for the counterexample.

We observe that $R(a) > r(a)$, $R(b) > r(b)$, and $R(c) > r(c)$, and also $K_R(v) > K_r(v)$. However, the conclusion of the Schwarz Lemma fails, as:

$$R(v) = 1.47981 < 1.48297 = r(v)$$

This demonstrates that the Discrete Schwarz Lemma does not hold for arbitrary inversive distances.

Metric r:	Vtx	Radius	Curvature	Metric R:	Vtx	Radius	Curvature
	a	1.0	2.4930		a	1.0919	1.2490
	b	1.0	4.5427		b	2.4579	5.2003
	c	1.0	3.9797		c	2.2206	4.4997
	v	1.4829	4.6925		v	1.4798	4.7589

Conclusion and Future Work

We have successfully extended the Discrete Schwarz Lemma to circle packings with intersection angles. This work opens up new directions for research in discrete geometry and complex analysis.